MSME Early Warning and Intervention System

Phase 0 Proposal

Team Name: Group_4

Team Members:

Team Member Name	Location
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A focused web application proposal for earlier identification, explanation, and follow up of MSME financial stress.

Purpose: define a practical, testable application requirement that a three person development team can build without real customer data or direct access to bank systems.

Proposal Summary

Small businesses can face cash flow pressure before a serious loan repayment issue is visible. Sales may fall, customers may pay invoices late, overdue receivables may rise, and use of working capital may increase. These signals may be seen separately, making it harder for a relationship manager to decide when to act.

The proposed web application brings three selected signals into one explainable workflow. It identifies early deterioration, explains the drivers, suggests an appropriate next step for review, and records whether the action helped the MSME stabilize.

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Table 1: Proposal Summary

Item	Proposal decision
Primary user	Bank Relationship Manager
Secondary users	MSME Representative and Risk Admin Officer
Prototype data	Synthetic financial snapshots supplied through a mock Financial Data Service
Core workflow	Detect -> Explain -> Recommend -> Act -> Track
Decision boundary	The system supports review; it never approves credit, restructures loans, or makes autonomous financial decisions.

1 Problem Definition

1.1 Problem being addressed

An MSME can look healthy while its financial position is getting worse. For example, customers may pay invoices late while the MSME must still pay suppliers, employees, rent, taxes, and loan obligations. The bank may see a late repayment only after the earlier warning signs have already become serious.

The problem is delayed identification and intervention of early MSME financial stress. The system will help a relationship manager identify meaningful deterioration from a small, selected set of financial indicators and take a documented next step earlier.

1.2 Who experiences the problem

Table 2: Users Affected and Expected Benefits

User or group	Problem experienced	Benefit from the proposed system
Relationship Manager	Manages many MSME customers and must interpret separate financial signals before deciding whether to contact a customer.	Gets a prioritised, explainable warning and a structured follow up case.
MSME Representative	May not recognise that late collections and rising credit use are	Sees the reason for a warning and can respond to the bank with relevant

	creating cash flow pressure.	information.
Risk Admin Officer	Needs visibility of high priority cases and consistent monitoring rules.	Can review escalations and manage the prototype rule settings.

1.3 How the process works today

Financial and credit information may be available in different places, such as invoices, cash flow records, working capital information, and repayment history. A relationship manager may need to look at these items separately, decide whether a pattern is temporary or serious, contact the MSME, and later remember to check the outcome. Existing systems can cover parts of this process, but this proposal focuses on the handoff from early warning to a recorded human intervention.

The proposal does not replace existing bank risk or customer-monitoring systems. It addresses the focused operational gap shown below.

Table 3: Current State, Focused Gap, and Proposed Improvement

Perspective	**Positioning**
Current State	Financial and credit indicators may already be available and monitored through existing banking processes and systems. Relationship Managers and banking teams may review these indicators as part of customer and risk management.
Focused Gap	The gap is not a lack of monitoring. It is consistently consolidating selected early-warning indicators, recognising meaningful combinations of deterioration, explaining why the MSME is being flagged now, converting the warning into a structured human-reviewed intervention, and tracking the outcome.
Proposed Improvement	The application validates selected financial data; monitors a small number of indicators; identifies explainable stress patterns; shows Why Now using previous-versus-current evidence; recommends a non-binding next action; requires a Relationship Manager decision; records the intervention and follow-up; and compares later data to show Improving/Stabilising, Still Stressed, or Escalated.

1.4 Impact and value

- For MSMEs: delayed action can turn a short term collection issue into a more serious cash flow problem.
- For relationship managers: manual review makes it harder to consistently prioritise the customers who need attention.
- For banks: later intervention can increase operational effort and the risk that a viable MSME becomes more financially stressed.
- For the prototype: the workflow makes deterioration visible, explainable, and testable without using confidential data.

1.5 Assumptions and exclusions

- The prototype uses synthetic financial snapshots. It does not use real customer information, bank credentials, or live banking data.
- A warning is a prompt for human review. It is not a credit decision, a default prediction, or a statement that the MSME will fail.
- The prototype assesses only revenue deterioration, overdue receivables, and credit utilisation.
- Recommendations are guidance for the relationship manager. The manager decides whether and how to act.

2. Users and Workflow

2.1 User roles and Actions

Role	Goal	Main actions
Relationship Manager	Ensure MSMEs requiring attention are reviewed, acted on, and followed by promptly.	Review assigned portfolio and warning evidence; accept, reject, or escalate warnings; record interventions and outcomes; set and track follow-up dates;
MSME Representative	Understand the warning, provide relevant information, and complete agreed follow-up actions.	View own risk/health status and explanation; acknowledge warnings; provide comments or updated information; review agreed actions and follow-up dates.
Risk Admin Officer	Maintain consistent monitoring and oversee serious or escalated risk cases.	Review high-priority cases and escalations; configure monitoring thresholds; review system-wide risk patterns; Close alerts with a documented reason when appropriate.

2.2 Primary end to end workflow

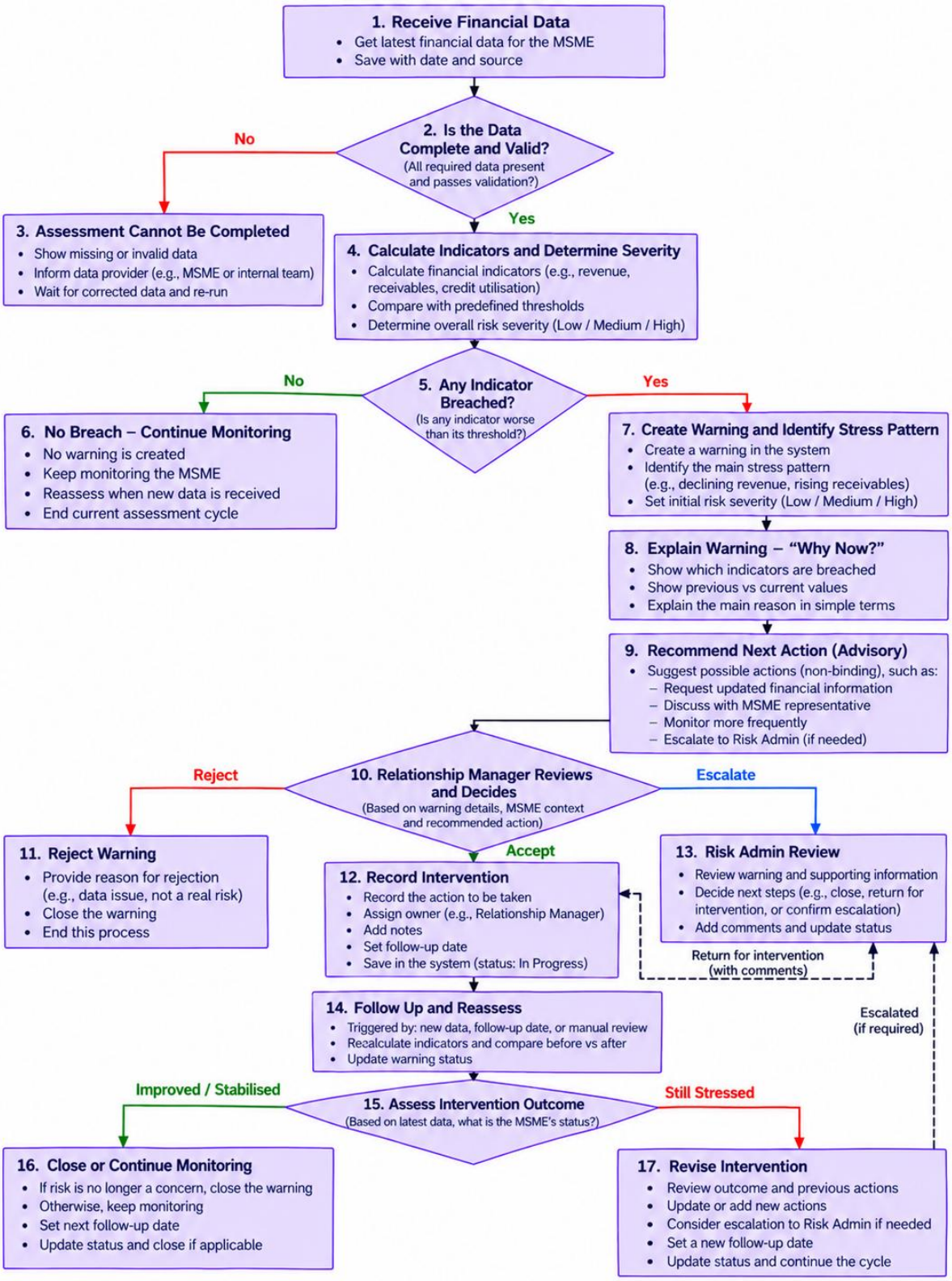
The workflow below represents the primary end-to-end business process for the application. It covers the normal processing path, warning scenarios, rejection paths, escalation paths, and failure conditions.

Closed-loop workflow:

Financial Data Received → Data Validation → Indicator and Severity Assessment → Warning Detection → "Why Now?" Explanation → Non-Binding Next-Action Recommendation → Relationship Manager Decision → Intervention and Follow-up → Updated Financial Data → Outcome Assessment

The outcome assessment compares financial indicators before and after the intervention. It shows the observed change in the MSME's financial condition and does not imply that intervention directly caused the outcome.

Figure 1: Primary End-to-End Early Warning, Human Intervention, and Outcome Assessment Workflow



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2.3 Main decisions and status changes

Table 5: Main Decisions and Status Changes

Event	Decision	Status after decision
Assessment completed	System calculates indicators, checks configured thresholds, and determines the current risk level.	Healthy, Watch, Early Stress, or High Priority.
Warning reviewed	Relationship Manager reviews the warning evidence and chooses to accept, reject, or escalate the warning.	Intervention Planned, Closed Reviewed, or Escalated.
Intervention logged	Relationship Manager records the agreed action, owner, notes, and follow-up date.	Intervention In Progress.
Follow up completed	Latest financial data is compared with the previous assessment to determine whether the MSME has improved, stabilized, or remains stressed.	Improved, Stabilised, Still Stressed, or Escalated.
Data validation failure	Required financial data is missing, invalid, or unavailable, so the assessment cannot be completed.	Assessment Unavailable.

3 Capabilities and Requirements

3.1 Core application capabilities

Table 6: Core Application Capabilities

ID	Capability	What the application must do
C1	MSME financial profile	Store each MSME's core profile, assigned Relationship Manager, approved credit limit, and dated financial snapshots required for assessment and monitoring.
C2	Financial health monitoring	Calculate defined financial indicators using the latest and previous financial snapshots, store the calculated values, and show how each indicator has changed over time.
C3	Early warning engine	Validate that all required assessment data is available, apply configured indicator thresholds and business rules, determine the risk severity, and create a warning when one or more

		warning conditions are met. If required data is missing or invalid, do not create a new assessment or warning and show the reason.
C4	Explanation and recommendation	For each warning, show the breached indicators, current and previous values, applicable thresholds, calculated severity, identified stress pattern, and a clear "Why Now?" explanation. Provide a suggested next action as an advisory recommendation for the Relationship Manager.
C5	Intervention and follow up tracking	Allow the Relationship Manager to accept, reject, or escalate a warning. For an accepted warning, record the agreed intervention, owner, notes, status, and follow-up date. When new financial data becomes available or the follow-up date is reached, support reassessment and record whether the MSME has improved, stabilized, or remains stressed.

3.2 Functional requirements and acceptance criteria

Table 7: Functional Requirements and Acceptance Criteria

Requirement	**Observable behaviour**	**Acceptance criteria**
FR1 Portfolio view	The Relationship Manager sees only assigned MSMEs with latest health status and data date.	Given assigned MSMEs, the list shows business name, severity, latest assessment date, and a visible stale-data marker where relevant.
FR2 Assessment	The system calculates each indicator from valid monthly snapshots.	Given three valid periods, the displayed formulas and calculated values match the stored data.

FR3 Warning	The system creates a warning with severity and evidence when one or more indicators breach a rule. It also identifies an explainable stress pattern and shows why the status changed now.	Given a change from Watch to Early Stress, the warning compares the relevant previous and current values, states the breached indicators, and gives a plain-language escalation explanation.
FR4 Recommendation	The system gives a non-binding action based on the main stress driver.	A receivables breach suggests overdue invoice review and possible invoice-financing discussion; it does not suggest automatic lending.
FR5 Intervention	The manager can accept, reject, or escalate a warning.	Accepting requires an action, owner, and follow up date. Rejecting requires a reason. Both decisions are saved in the case history.
FR6 Follow up	The manager can record an outcome after follow-up data is available. The system compares relevant indicators before and after the intervention.	The case shows Improving/Stabilising, Still Stressed, or Escalated with the before and after values. It states that indicators improved after intervention; it does not claim the intervention caused the improvement.

3.3 Prototype business rules and formulas

The values below are initial prototype assumptions. They are configurable business-rule parameters, not RBI-mandated thresholds. They are used so the workflow is observable and testable. A production system would calibrate them using bank policy, historical data, and domain validation.

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Table 8: Prototype Business Rules and Formulas

Rule	**Formula**	**Prototype default**	**System behaviour**
BR1 Revenue deterioration	Revenue decline % = ((previous period revenue - current period revenue) / previous period revenue) x 100	At least 15% decline compared with the immediately preceding period for two consecutive periods.	Mark revenue indicator as breached after the second consecutive qualifying decline.
BR2 Overdue receivables	Overdue receivables % = (overdue receivables / total receivables) x 100	More than 30% of total receivables.	Mark receivables indicator as breached for the current assessment.
BR3 Credit utilisation	Credit utilisation % = (credit used / sanctioned credit limit) x 100	More than 70% of the sanctioned credit limit.	Mark credit indicator as breached for the current assessment.
BR4 Severity	Breached indicator count = number of BR1, BR2, and BR3 that are breached.	0 = Healthy; 1 = Watch; 2 = Early Stress; 3 = High Priority.	Create or update the assessment severity using the count.
BR5 Data completeness	Required values: revenue for current and two prior periods, total and overdue receivables, credit used, sanctioned limit.	All required values present; denominators greater than zero.	If incomplete, show Assessment Unavailable and do not create a new warning.

BR6 Stress pattern	Evaluate combinations of BR1, BR2, and BR3 after severity is calculated.	Use the three prototype pattern rules; Broad Financial Deterioration has precedence.	Show one explainable dominant pattern when a defined combination is met. The pattern does not change severity or make a financial decision.
BR7 Outcome assessment	Compare the latest valid snapshot with the snapshot used before intervention.	Improving/Stabilising: severity is lower and at least one relevant indicator improves. Still Stressed: severity remains the same or a material indicator remains breached. Escalated: severity rises or the case is escalated by a human reviewer.	Show before and after values, outcome, and data date. State that indicators improved after intervention; do not state that intervention caused the improvement.

Stress Pattern Classification is a lightweight, rule-based explanation layer. It does not change severity and is not a diagnosis of financial distress, a default prediction, or an automatic credit decision.

Table 9: Stress Pattern Classification Rules

Prototype pattern	**Rule-based condition**	**Plain-language explanation**
Collection/Liquidity Stress	Overdue receivables is breached and either revenue deterioration or high credit utilisation is also breached.	Delayed collections appear to be creating cash flow pressure. The MSME may have money due from customers but less cash available for current obligations.
Credit Dependency Stress	Revenue deterioration and high credit utilisation are both breached.	Operating performance is weakening while dependence on credit is high. A Relationship Manager review is needed; this is not an automatic lending decision.
Broad Financial Deterioration	Revenue deterioration, overdue receivables, and credit utilisation are all breached.	Several major indicators are deteriorating at the same time. Treat this as a High Priority review condition, not as an automatic default or credit decision.

Pattern precedence: Broad Financial Deterioration takes precedence when all three indicators are breached. Otherwise, Collection/Liquidity Stress takes precedence when overdue receivables are involved. A Watch case with only one breached indicator may have no pattern classification.

Validation rules: monetary values cannot be negative; overdue receivables cannot exceed total receivables; credit used cannot be negative; sanctioned credit limit must be greater than zero; only one financial snapshot can exist for an MSME and reporting period; dates must be in chronological order.

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3.4 Recommendation mapping

Table 10: Recommendation Mapping

Primary driver	Suggested next action for review	Decision boundary
Revenue deterioration	Discuss the sales trend, key customers, and whether a business review is needed.	No automatic credit decision.
Overdue receivables	Review overdue invoices, identify delayed buyers, and discuss whether eligible invoice-financing options should be explored.	The prototype does not execute TReDS or invoice financing.
Credit utilisation	Review current working capital use and upcoming obligations with the MSME.	The prototype does not approve, change, or disburse a credit facility.
Multiple drivers	Arrange a Relationship Manager review and consider escalation to Risk Admin.	Human review is mandatory for escalation.

3.5 Failure and exception scenarios

Table 11: Failure and Exception Scenarios

Scenario	Expected application behaviour	User outcome
Missing or invalid financial data	Validate the snapshot. If required data is missing or invalid, reject the import or mark the assessment unavailable.	User sees which fields require correction; no misleading new health level is shown.
Financial Data Service unavailable	Show the last successful snapshot date and a stale-data warning. Do not run a new assessment from partial data.	Relationship Manager can view prior information but knows that it is not current.
Duplicate reporting period	Reject a second snapshot for the same MSME and period unless an authorised correction workflow is used.	Existing record is protected and the user receives a clear duplicate message.
Warning disputed or rejected	Require the Relationship Manager to select or enter a reason. Preserve the warning and review history.	Case becomes Closed Reviewed; the decision is auditable.
Recommendation cannot be completed	Allow the manager to record the reason, select an alternative action, and set a revised follow up date.	Case stays Intervention In Progress or is escalated; it is not silently closed.

4 Data and External Interfaces

4.1 Business entities and relationships

The application stores business workflow data, not only user accounts. One MSME has many financial snapshots and assessments. An assessment can create one warning. A warning can have several intervention and follow up records. Each MSME is assigned to one Relationship Manager for the prototype.

Figure 2: MSME Early Warning System Data Model and Entity Relationships

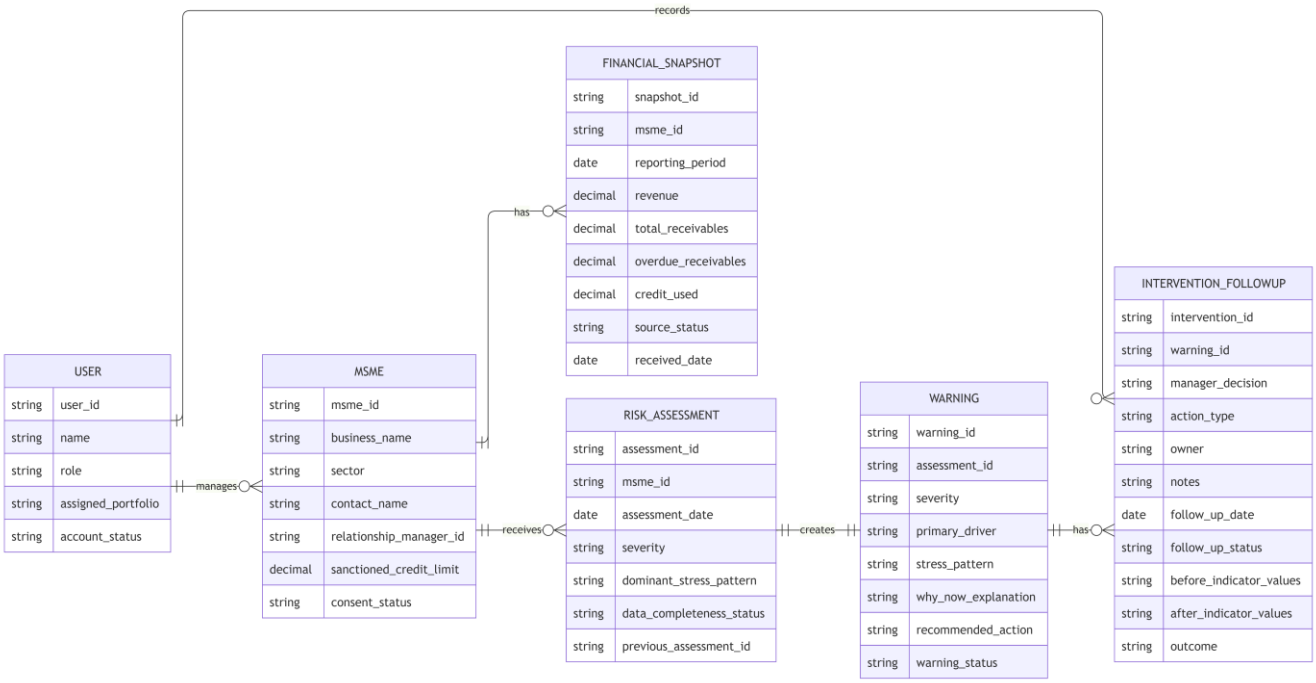


Table 12: Business Entities and Key Fields

Entity	**Key fields**
MSME	MSME ID, business name, sector, contact name, assigned Relationship Manager, sanctioned credit limit, consent status.
Financial Snapshot	Snapshot ID, MSME ID, reporting period, revenue, total receivables, overdue receivables, credit used, source status, received date.
Risk Assessment	Assessment ID, MSME ID, assessment date, each indicator result, severity, dominant stress pattern, data completeness, calculation inputs, previous assessment reference.
Warning	Warning ID, assessment ID, severity, primary driver, stress pattern, Why Now comparison, explanation, recommended action, warning status.
Intervention	Intervention ID, warning ID, decision, action type, owner, notes, due date, follow up status, before and after indicator values, outcome.
User	User ID, name, role, assigned portfolio where applicable, account status.

4.2 External interface and prototype simulation

In a production setting, selected financial data would normally come from approved bank or MSME accounting sources with proper consent and credentials. Those interfaces are confidential or unsuitable for this training prototype. Therefore, the prototype will use a narrow mock Financial Data Service. It reproduces only the response behaviour needed by the application. It does not reproduce a bank, payment network, credit bureau, or accounting system.

Table 13: Mock Financial Data Service Requirements

Interface detail	**Prototype requirement**
Interface name	Mock Financial Data Service
Purpose	Provide monthly financial snapshot data for a selected MSME.
Availability	Local mock API or static JSON fixture. No registration, authentication, or live customer data is required for the prototype.
Fallback	If the service fails, retain the last successful snapshot and mark the data as stale. The user may retry the request.
Data source	Synthetic scenarios designed to show healthy, watch, early stress, high priority, and stabilising paths.

4.3 Sample interface contract

Sample request

```
GET /api/msmes/MSME-1001/financial-summary?period=2026-03
```

Successful response

```
{
 "msmeId": "MSME-1001",
 "period": "2026-03",
 "revenue": 1600000,
 "totalReceivables": 1000000,
 "overdueReceivables": 380000,
 "creditUsed": 720000,
 "sanctionedCreditLimit": 1000000,
 "sourceUpdatedAt": "2026-04-01T09:00:00Z"
}
```

Failure response

```
{
 "errorCode": "FINANCIAL_DATA_UNAVAILABLE",
 "message": "Financial summary is temporarily unavailable."
}
```

Expected application behaviour: on success, validate and store the snapshot, then run the assessment. On failure, show the error, retain the last successful data date, mark the assessment as stale, and do not create a new severity from missing data.

5 Scope Risks and Future Enhancements

5.1 Included in the prototype

- Three user roles: Relationship Manager, MSME Representative, and Risk Admin Officer.
- Synthetic MSME profiles and monthly financial snapshots.
- Three configurable indicators, severity calculation, stress pattern classification, Why Now explanations, and recommendation mapping.
- Portfolio view, alert detail, review decision, intervention record, follow up, before and after comparison, and outcome tracking.
- Mock Financial Data Service with success, duplicate, invalid-data, and unavailable-data scenarios.

5.2 Not included

- Live bank, payment, GST, accounting, TReDS, or credit-bureau integration.
- Real customer information, banking credentials, or confidential financial data.
- Loan application, approval, pricing, disbursement, restructuring execution, or collections execution.
- Bankruptcy or default prediction, automated risk decisions, or autonomous recommendations that change a customer account.
- A full accounting system, enterprise resource planning system, or replacement for existing bank risk systems.

5.3 Dependencies and key risks

Table 14: Dependencies and Key Risks

Risk or dependency	**Why it matters**	**Prototype mitigation**
Synthetic data quality	Unrealistic values could create an unconvincing demo.	Use named scenarios with internally consistent trends and test every expected severity.
False interpretation of a warning	A warning may be treated as a lending or default decision.	Use clear wording: review prompt only; show evidence and require a human decision.
Threshold suitability	Prototype defaults may not suit every sector or bank.	Make thresholds configurable and label them as prototype assumptions.
Data freshness	Old data can make a case look safer or riskier than it is.	Display source timestamp and stale-data status; block new assessment on a failed fetch.
Scope growth	Adding live integrations or complex ML could make the project too large.	Keep the build limited to five capabilities and a mock interface.

5.4 Five future enhancements

Enhancement	**Future value**
Approved accounting data connection	With consent and approved access, import selected invoice and receivable information from an accounting system.
Bank policy calibration	Calibrate severity rules by sector, customer size, bank policy, and validated historical outcomes.
TReDS eligibility referral	Provide a guided referral or link-out after human review when an invoice-financing route may be appropriate.
Predictive analytics with governance	Use validated historical data to estimate a future deterioration risk, with explanations, monitoring, and human oversight.
Portfolio and intervention reporting	Show trend reports such as warning volumes, average time to review, intervention completion, and stabilisation outcomes.

7 Task and Contribution Register

Task	Person responsible	Output or evidence
Defined the MSME financial-stress problem, affected users, current-state gap, scope assumptions and exclusions.	Chandana K P	Problem definition; current-state gap; scope section.
Mapped the Relationship Manager, MSME Representative and Risk Admin Officer roles, including permitted decisions and responsibilities.	Chandana K P	User-role table; role explanations.
Designed the end-to-end workflow, decision points, status changes, normal path, warning path, rejection path and escalation path.	Yuva Prasath	Workflow diagram; main decisions and status-changes table.
Defined the three financial indicators, prototype thresholds, severity calculation, stress-pattern rules and validation rules.	Yuva Prasath	Business-rules tables; test scenarios.
Defined core capabilities, functional requirements, acceptance criteria, failure scenarios, data entities and mock Financial Data Service contract.	Brijesh Naik	Requirements tables; data model; mock API request and responses.
Reviewed the proposal for completeness, prepared the walkthrough explanation and validated the healthy, warning, rejected, unavailable-data and stabilising scenarios.	Brijesh Naik	Review checklist; walkthrough script; scenario evidence.